

Customer Shopping Behavior Analysis

A data-driven exploration of 3,900 transactions — uncovering spending patterns, customer segments, product preferences, and subscription behavior to guide strategic business decisions.

3,900

Total transactions analyzed

18

Dataset features

\$233,081

Total revenue captured

SQL + Python

Analytical stack

Project Overview

This project analyzes customer shopping behavior using transactional data from **3,900 purchases** across various product categories. The goal is to uncover insights into spending patterns, customer segments, product preferences, and subscription behavior to guide strategic business decisions.

The analysis follows a structured pipeline — from raw data ingestion and cleaning in Python, through SQL-based querying in PostgreSQL, to actionable business recommendations.

Analytical Objectives

- Spending Patterns
Understand how customers allocate spend across categories, genders, and seasons.
- Customer Segments
Identify high-value, discount-driven, and subscription cohorts.
- Product Preferences
Surface top-rated and discount-dependent products.

Dataset Summary

3,900

Total Rows

Individual purchase transactions

18

Columns

Feature dimensions per record

37

Missing Values

In Review Rating column only

Key Feature Groups

Customer Demographics

Age, Gender, Location, Subscription
Status

Purchase Details

Item Purchased, Category, Purchase
Amount, Season, Size, Color

Shopping Behavior

Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases,
Review Rating, Shipping Type

Data Preparation & Cleaning

The analytical pipeline began with systematic data preparation in Python before loading into PostgreSQL for SQL analysis.



Data Loading

Imported the dataset using `pandas` for initial inspection.



Initial Exploration

Used `df.info()` to check structure and `.describe()` for summary statistics.



Missing Data Handling

Imputed 37 missing values in Review Rating using the **median rating per product category**.



Column Standardization

Renamed all columns to `snake_case` for readability and documentation consistency.



Feature Engineering

Created `age_group` via age binning and `purchase_frequency_days` from purchase timestamps.



Database Integration

Verified discount/promo redundancy, dropped `promo_code_used`, then loaded cleaned DataFrame into PostgreSQL.

Revenue by Gender

Gender-segmented revenue analysis reveals a significant disparity in total spend between male and female customers. Male customers account for **more than twice** the revenue generated by female customers, suggesting either a larger male customer base or higher average transaction values among male shoppers.

\$157,890

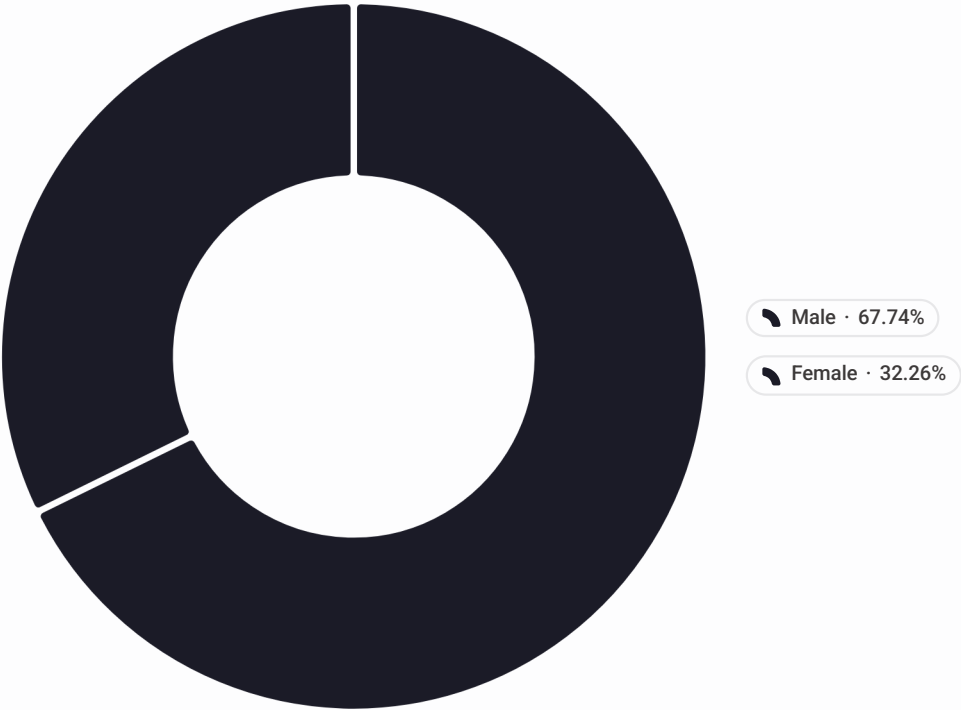
Male Revenue

67.7% of total revenue

\$75,191

Female Revenue

32.3% of total revenue



High-Spending Discount Users

839

Customers Identified

Used discounts yet spent above average purchase amount

Purchase Range

Transactions in this segment ranged from **\$62 to \$97** – well above the dataset average – despite discount application.

Segment Significance

This cohort of **839 customers** represents a strategically valuable segment: they are discount-responsive yet not discount-dependent. Their willingness to spend above average even when using discounts suggests strong underlying purchase intent and brand affinity.

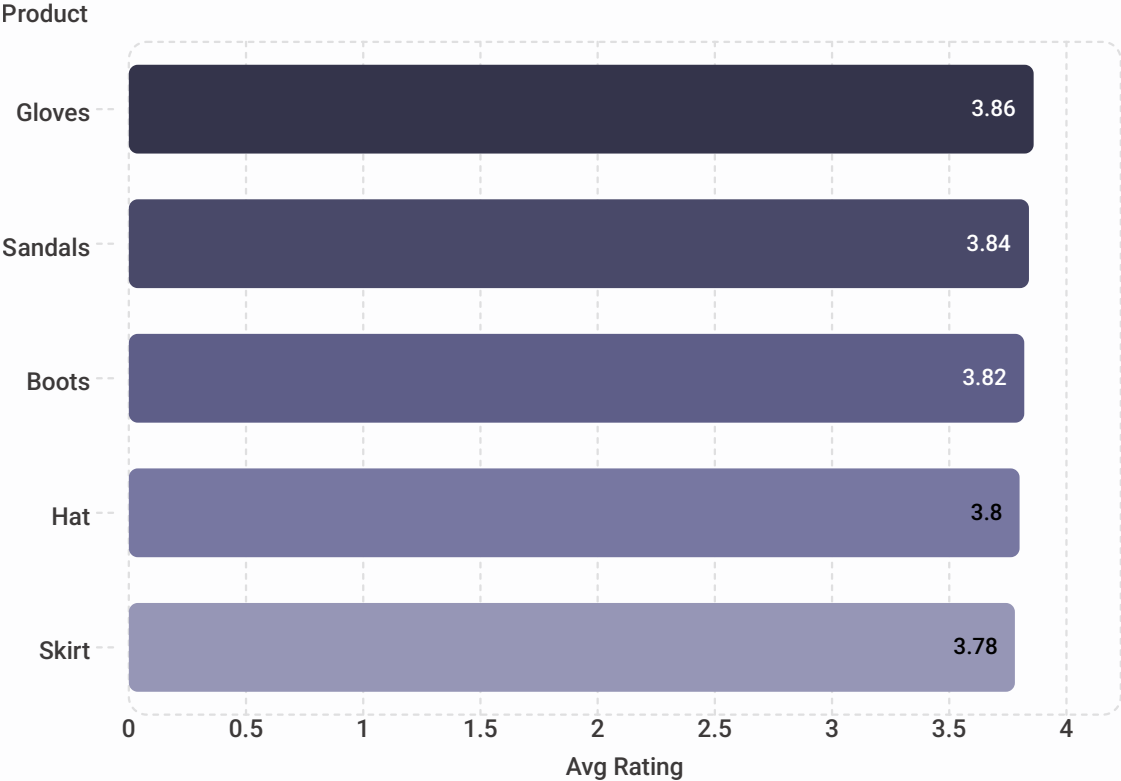
Retention Opportunity

These customers are ideal candidates for loyalty programs – they respond to incentives without requiring deep discounts to convert.

Upsell Potential

With spend already above average, targeted upsell and cross-sell campaigns could further increase basket size.

Top 5 Products by Rating



Rating Analysis

The top five rated products span accessories and footwear categories. Ratings are tightly clustered between **3.78** and **3.86**, indicating consistently positive but not exceptional customer satisfaction across the product range.

Product	Avg Rating	Category
Gloves	3.86	Accessories
Sandals	3.84	Footwear
Boots	3.82	Footwear
Hat	3.80	Accessories
Skirt	3.78	Apparel

Shipping Type Comparison & Subscriber Analysis

Shipping Type vs. Average Spend

Express shipping customers spend marginally more on average than Standard shipping customers — a **\$2.02 differential** — suggesting shipping preference is not a strong predictor of purchase value.

Standard Shipping
\$58.46 avg purchase amount
Express Shipping
\$60.48 avg purchase amount

Subscribers vs. Non-Subscribers

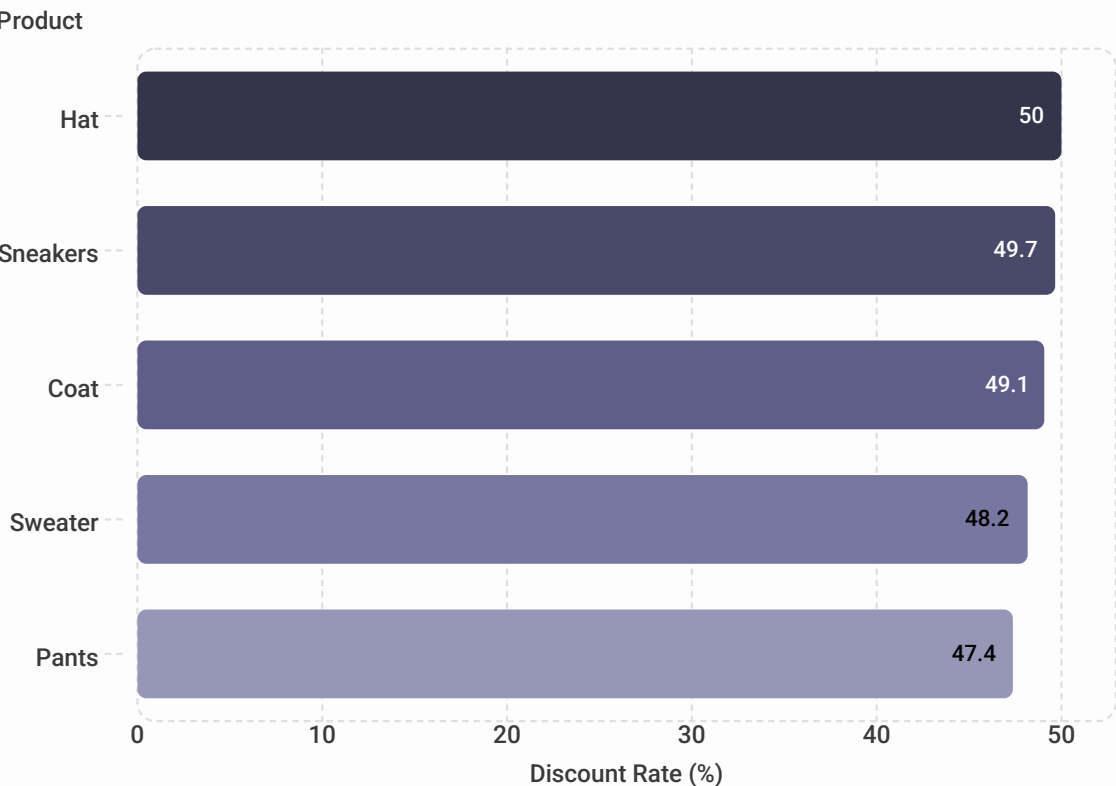
Non-subscribers vastly outnumber subscribers and drive the majority of total revenue, though average spend per transaction is nearly identical between the two groups.

Segment	Customers	Avg Spend	Total Revenue
Subscribers	1,053	\$59.49	\$62,645
Non-Subscribers	2,847	\$59.87	\$170,436

❑ Non-subscribers represent **73% of customers** and **73.1% of total revenue** — a significant conversion opportunity.

Discount-Dependent Products

The following products exhibit the highest rates of discount application, indicating potential over-reliance on promotional pricing to drive sales volume.



Strategic Implications

All five products show discount rates approaching or exceeding **50%** – meaning nearly half of all purchases for these items involve a discount. This pattern warrants careful review:

Margin Risk

Persistent discounting at ~50% rates erodes margin and may condition customers to wait for promotions before purchasing.

Pricing Review

Hat leads at exactly 50.00% – a signal that list price may be misaligned with perceived value, requiring a pricing strategy reassessment.

Promotional Discipline

Reducing discount frequency on high-rated products (Hat also appears in Top 5 by rating) could preserve revenue without sacrificing demand.

Key Findings & Strategic Recommendations

Gender Revenue Gap

Male customers generate **\$157,890** vs. female **\$75,191**. Investigate whether this reflects audience size or spend depth — and develop targeted campaigns to grow the female segment.

High-Value Discount Users

839 customers spend above average even with discounts (\$62–\$97 range). Prioritize this cohort for loyalty and upsell programs — they convert without deep incentives.

Subscription Conversion

Only **1,053 of 3,900** customers are subscribers. With near-identical avg spend (\$59.49 vs. \$59.87), converting non-subscribers could significantly boost recurring revenue.

Discount Discipline

Hat, Sneakers, Coat, Sweater, and Pants all carry ~50% discount rates. Implement promotional guardrails to protect margin on these high-volume SKUs.

Product Quality Signal

Top-rated products (Gloves 3.86, Sandals 3.84, Boots 3.82) cluster in accessories and footwear. Amplify marketing investment behind these proven performers to drive organic growth.

Shipping Insight

The \$2.02 gap between Express (\$60.48) and Standard (\$58.46) avg spend is minimal. Shipping preference is not a reliable proxy for customer value — segment by spend behavior instead.

Data Infrastructure

The Python → PostgreSQL pipeline established here provides a scalable foundation for ongoing analysis. Recommend scheduling automated refreshes and building a monitoring dashboard for key KPIs.